

This document explains how to set up, install dependencies, and run the ESP32 firmware and MATLAB script associated with the wearable bruxism/TMJ monitoring system.

I. Overview

The project consists of two components:

- ESP32 firmware (written in Arduino C++)
 - Connects to Wi-Fi, samples multiple sensors (EMG, contact microphone, thermistor, accelerometer), processes the data, and serves it via a web interface and API.
- MATLAB visualization script
 - Connects to the ESP32 over HTTP, filters and plots data in real time, and detects threshold-based events.

II. Required Hardware

- ESP32-WROOM-32 microcontroller (or equivalent)
- EMG sensor (MyoWare)
- Contact microphone
- Thermistor (NTC)
- Accelerometer (ADXL326)
- USB cable for programming the ESP32

III. Software Requirements

Arduino IDE (for ESP32):

- Install the latest version from <https://www.arduino.cc/en/software>
- Install the ESP32 board definitions:
 - Open Arduino IDE.
 - Go to Tools > Board > Boards Manager.
 - Search for ESP32 and install the package by Espressif Systems.

Required Arduino Libraries:

- WiFi.h (included in the ESP32 board package)
- AsyncTCP

- ESPAsyncWebServer

If not available through the Library Manager:

- Download from:
 - <https://github.com/me-no-dev/AsyncTCP>
 - <https://github.com/me-no-dev/ESPAsyncWebServer>
- In Arduino IDE, go to Sketch > Include Library > Add .ZIP Library
- Import the downloaded ZIPs

MATLAB

Ensure that MATLAB is installed with the following toolboxes:

- Signal Processing Toolbox
- Base MATLAB (includes webread, webwrite, plot, etc.)

No additional add-ons are required.

IV. How to Run the System

Step 1: Upload Code to ESP32

1. Open the Arduino IDE.
2. Open the ESP32 code file (Final_Script_Senior_Design.ino).
3. Replace the Wi-Fi SSID and password in the code with your own.
4. Connect your ESP32 via USB.
5. Select your board and port under Tools > Board and Tools > Port.
6. Click Upload.
7. Open the Serial Monitor at 115200 baud to confirm that the ESP32 connects to Wi-Fi and displays its IP address.

Step 2: View Real-Time Sensor Web Interface

1. **On a separate device connected to the same Wi-Fi network**, open a browser.
2. Enter the IP address shown in the Serial Monitor.
3. A webpage will display live values for EMG, contact mic, orientation, wear status, and event count.

Step 3: Run MATLAB Visualization

1. Open MATLAB.
2. Open the script `real_time_plot.m`.
3. Edit the `esp32_ip` variable at the top of the script to match the IP address shown by your ESP32.
4. Run the script.
5. You will see:
 - A real-time filtered signal plot
 - A frequency-domain FFT plot

V. Notes and Troubleshooting

- Ensure that your computer and the ESP32 are on the same Wi-Fi network.
 - Make sure you are using a personal/home Wi-Fi connection. Academic institutions will often block external connections.
- If the MATLAB script fails to connect, double-check the IP address and ensure the ESP32 is running.
- If you are using a new computer, verify that all the Arduino libraries and toolboxes mentioned above are installed.